

CS1003

Programming in C

Preliminaries

Computer

- A machine that can be programmed to carry out sequences of computations automatically.
- Store, retrieve, and process information or data – understands only 0s and 1s.
- Designed specifically to handle and process large amount of information quickly and efficiently.
- There exists several categorization of computers based on their architecture, size, purpose, etc.



ENIAC



IBM System 360
(Mainframe)



Laptop



Desktop



Supercomputer



Smartphone

Hardware and Software

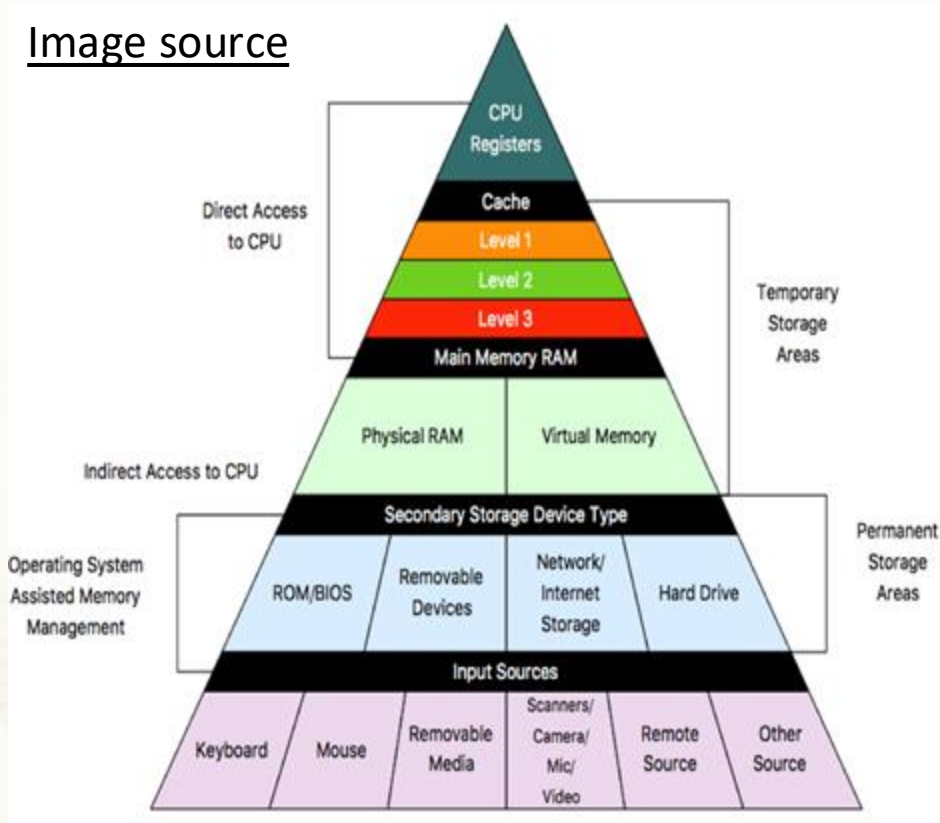
- Hardware: Physical parts of a computer. It is hard with respect to changes.
 - Central processing unit (CPU), random access memory (RAM), motherboard, computer data storage, graphics card, sound card, computer case, etc.
 - External devices such as a monitor, mouse, keyboard, and speakers.
- Software: Set of instructions that can be stored and run by hardware. It is soft as it is easy to change.
- Hardware gets direction from software to execute any command or instruction.

Types of Software

- Operating System Software (Microsoft Windows, macOS, Linux, Android, iOS, etc.)
- Application Software (Microsoft Office Suite, Adobe Photoshop, Spotify, Zoom, etc.)
- Utility Software (Antivirus software, Disk Cleanup, Backup Software, etc.)
- Programming Software (Integrated Development Environments – IDEs, like Visual Studio, Eclipse, etc.)
- System Software (Device drivers, Firmware, etc.)
- Embedded Software (used in smart devices)
- Open Source Software (Linux Kernel, Mozilla Firefox, LibreOffice, etc.)
- Proprietary Software (Microsoft Windows, Adobe Creative Suite, etc.)
- Freeware (Cleaner, Avast Free Antivirus, etc.)
- Shareware (WinRAR, WinZip, etc.)

CPU Arrangement

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Black lines indicate dataflow.
Red lines indicate control flow.

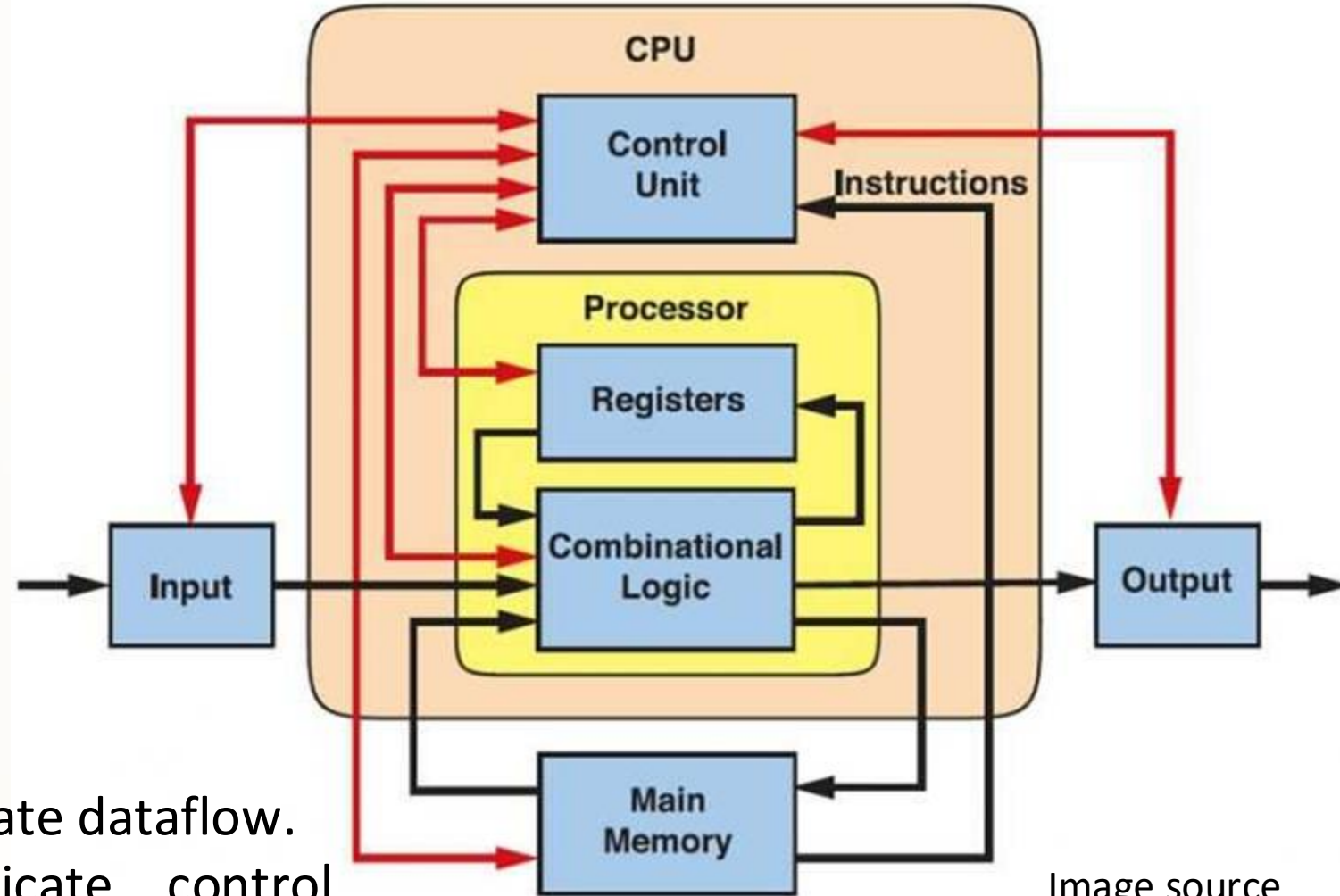


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Primary and Secondary

PRIMARY

- Hold program instructions, data, intermediate and final results of running programs.
- Faster operation.
- Smaller in size.
- Expensive.
- Loses data without power, i.e., its temporary.
- Example: RAM, ROM, etc.

SECONDARY

- Hold stored program instructions.
- Slower operation.
- Larger capacity.
- Cheaper.
- Retains data without power, i.e., its permanent.
- Example: hard disk, floppy, etc.

Why C???

- It doesn't seem to have an expiration date, because of the following features:
 - Closeness to the hardware,
 - Portability,
 - Deterministic usage of resources.
 - Versatility, efficiency and good performance
- Ideal for
 - Low level development (operating system kernels, embedded software, etc.)
 - High complexity data manipulation software, like databases or 3D animation.
- C is still unsurpassed when performance is the priority.



Contd...

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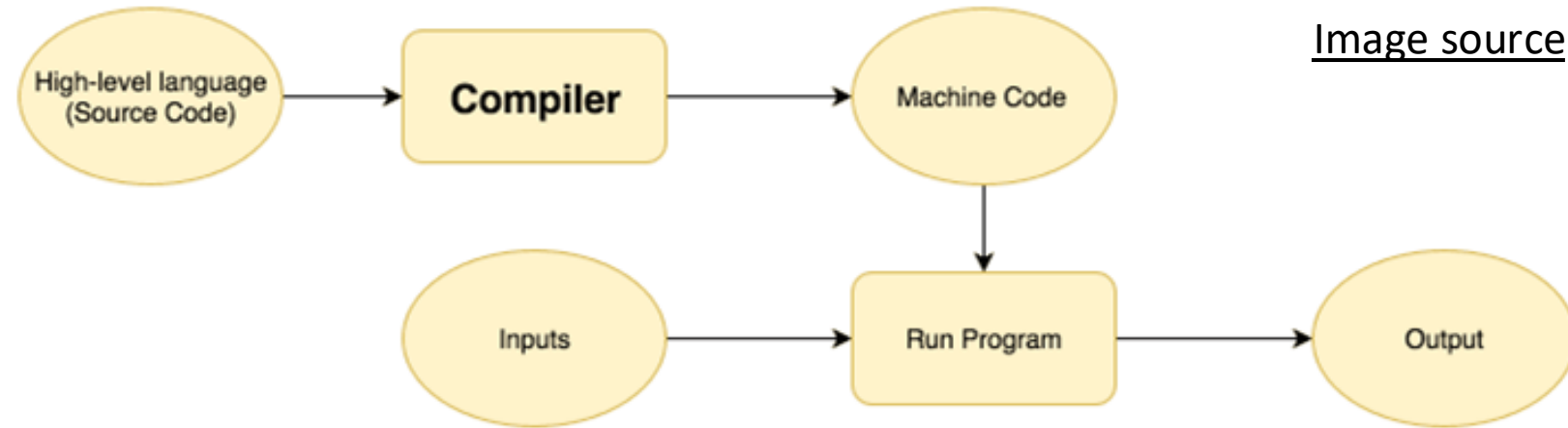
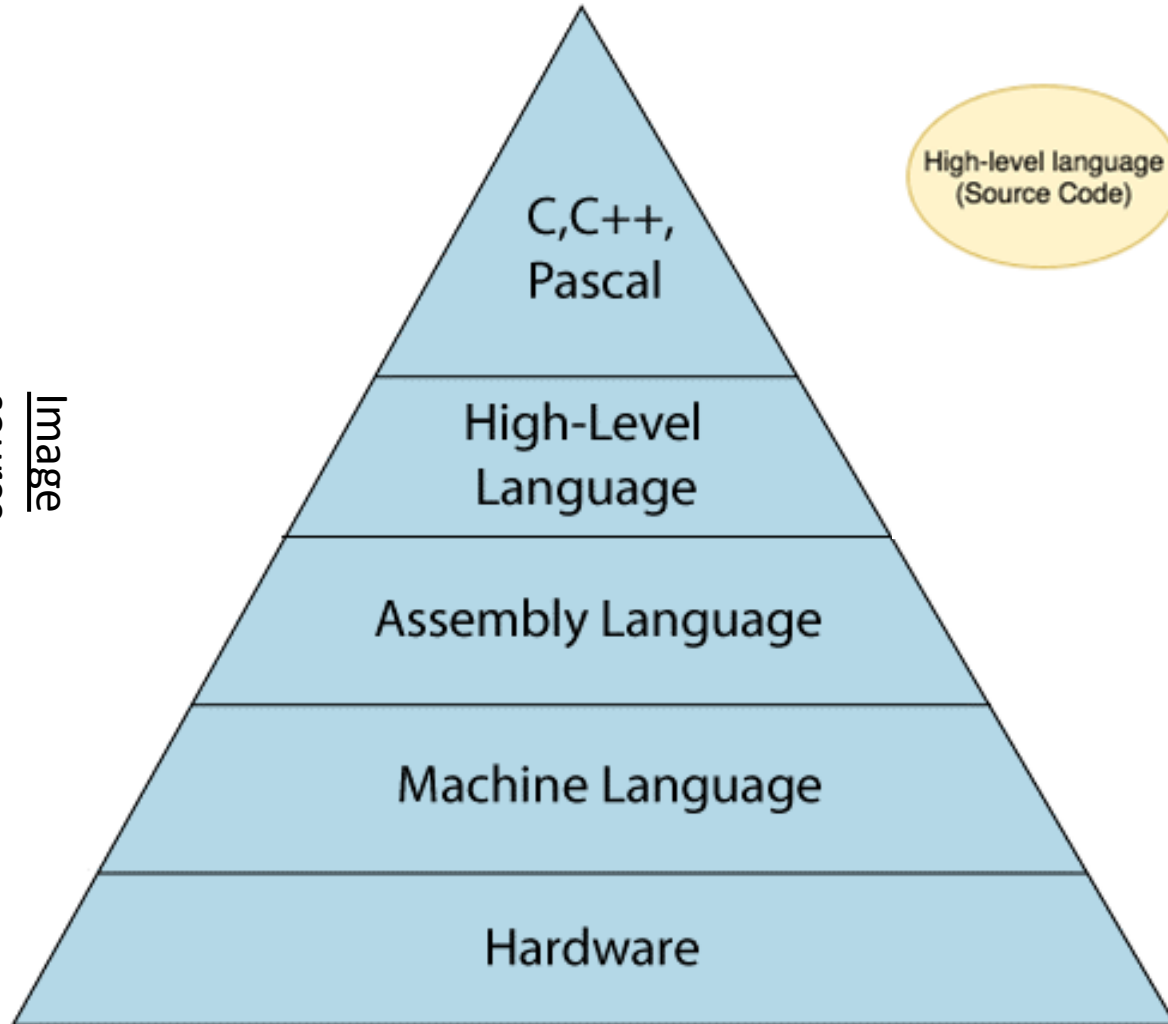


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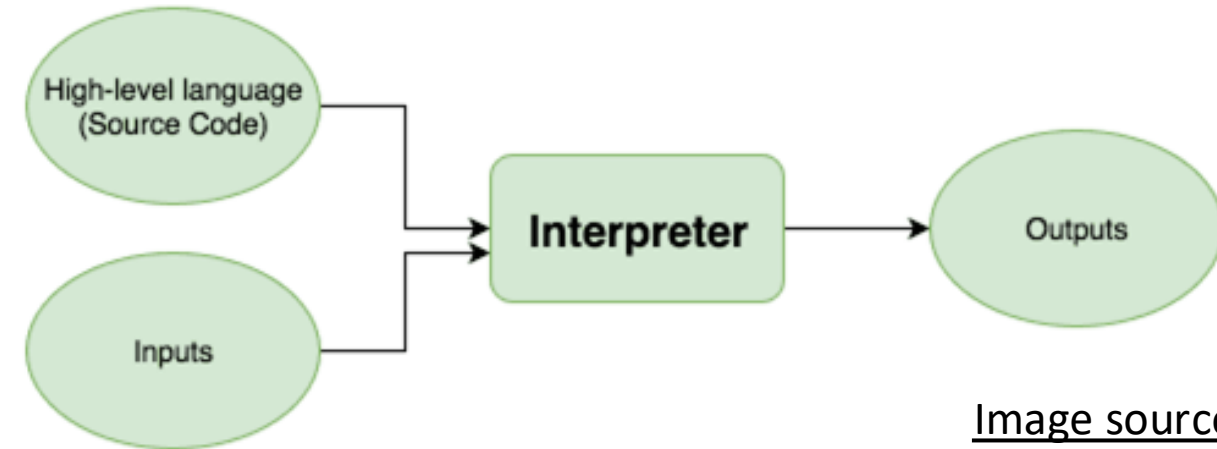


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Programming Language

Generation	First	Second	Third	Fourth	Fifth
Code example	1010101001100010 1001101010000001 1111111110100010	LDA 34 ADD #1 STO 34	$x = x + 1$	body.top { color : red; font-style : italic;}	Machines can take decisions
Language	Machine language	Assembly language	C, C++, Java, Visual Basic, JavaScript, etc.	Perl, Python, Ruby, SQL, MatLab, etc.	Mercury, OPS5, Prolog, etc.
General relation to object code	-	One to one	One to many	One to many	One to many

Programming Language

- Language designed to write instructions that can be executed by a computer.
- It has:
 - Set of words or **lexical** or **keywords**.
 - Set of rules or **syntax**.

Bits and Bytes

- A "bit" is atomic: the smallest unit of storage
- One bit stores two patterns (0 or 1); Two bits store four patterns (00 or 01 or 10 or 11), ..., n bits store 2^n patterns.
- One byte = collection of 8 bits. It is a unit of information storage.
- 1 Kilo Byte (KB) = 1024 or 2^{10} bytes.
- 1 Mega Byte (MB) = 1024 KB
- 1 Giga Byte (GB) = 1024 MB
- 1 Tera Byte (TB) = 1024 GB
- Peta Byte Exa Byte Zetta Byte Yotta Byte Bronto Byte Gonn Byte

Decimal to Binary

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Thank You